



ISLAMIC UNIVERSITY OF TECHNOLOGY

Lab Report 2

Livestock Management System

CSE 4408: System Analysis and Design Lab

Lab 2: Understanding and Modeling Organizational Systems

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Organizational Context and System Scope

Organizational Unit Overview

The subject of this system development effort is an Integrated Smart Farm Organization that operates across three primary livestock subsystems: Poultry Farming, Fisheries Management, and Cattle Rearing. The core function of this organization is to produce, manage, and distribute livestock-related products including eggs, poultry meat, fish, dairy, and cattle products in a sustainable and efficient manner. The farm employs farm workers, veterinarians, and managers under the strategic oversight of the farm owner, and engages with external parties including feed suppliers, buyers/merchants, and veterinary services.

The organization currently operates under a predominantly manual and paper-based model, where records for feed consumption, animal health, waste transfer, and inventory are maintained by hand. This operational mode, while familiar to the workforce, introduces significant inefficiencies, data inconsistencies, and delayed decision-making that directly impede the farm's capacity for growth and sustainability.

Core Problem Statement

During Lab 1, the following core problems were identified within the existing operations of the integrated farm:

- **Manual Record-Keeping and Data Loss:** All transactions—feed logs, health records, and inventory updates—are tracked on paper, leading to frequent data loss, duplication, and inaccuracies that compromise decision quality.
- **Disconnected Subsystems:** The poultry, fisheries, and cattle subsystems currently operate in silos with no shared data infrastructure, preventing resource optimization across the farm (e.g., using poultry waste as fish feed).
- **Poor Feed and Waste Tracking:** There is no systematic mechanism for tracking feed distribution quantities, schedules, or waste outputs, resulting in overfeeding, underfeeding, and missed resource-sharing opportunities.

- **Delayed Decision-Making:** The absence of real-time data means that farm managers and owners must wait for periodic manual reports before acting on health alerts, inventory shortfalls, or production anomalies.
- **No Real-Time Monitoring:** There is no centralized dashboard or monitoring capability to track animal health indicators, environmental conditions, or production metrics across subsystems simultaneously.

System Boundary Definition

The following table provides a clear delineation of what falls within and outside the proposed Livestock Management System boundary.

IN SCOPE	OUT OF SCOPE
Automated tracking of feed distribution and waste transfers between subsystems	Transport and GPS tracking of delivery vehicles and logistics
Real-time health monitoring and veterinary record management per batch	Integration with physical IoT devices, sensors, or automation hardware
Inventory management for feed stock, medicines, and farm supplies	Integration with external marketplace or e-commerce platforms
Centralized reporting and analytics dashboards for all subsystems	Financial accounting, payroll processing, or ERP-level financial modules
Livestock batch lifecycle tracking (arrival, growth, health, sale)	Customer-facing portals or mobile apps for buyers and merchants

IN SCOPE	OUT OF SCOPE
Worker task scheduling and daily activity logging	Government regulatory compliance reporting or export documentation
Supplier order management and stock replenishment requests	Cross-farm or multi-organization data sharing networks

Table 1: System Scope — In Scope vs. Out of Scope

Boundary Rationale

The defined system boundary focuses exclusively on the internal operational and informational needs of the farm organization. By centralizing feed tracking, health monitoring, inventory management, and worker scheduling within a single platform, the system directly addresses the five core problems identified in Lab 1. Excluding hardware automation and marketplace integration keeps the scope feasible within a software-only solution while still delivering substantial operational value. The boundary ensures that the most impactful pain points—data fragmentation, manual logging, and delayed decisions—are resolved without overextending development scope.

Context-Level Data Flow Diagram

Overview

The Context-Level Data Flow Diagram (DFD) provides a bird's-eye view of the Livestock Management System, illustrating how it interacts with external entities without revealing internal processing details. At this level, the entire system is represented as a single process (Process 0), and all actors that send data to or receive data from the system are identified as external entities. The diagram captures the primary data flows across all major stakeholders.

Diagram

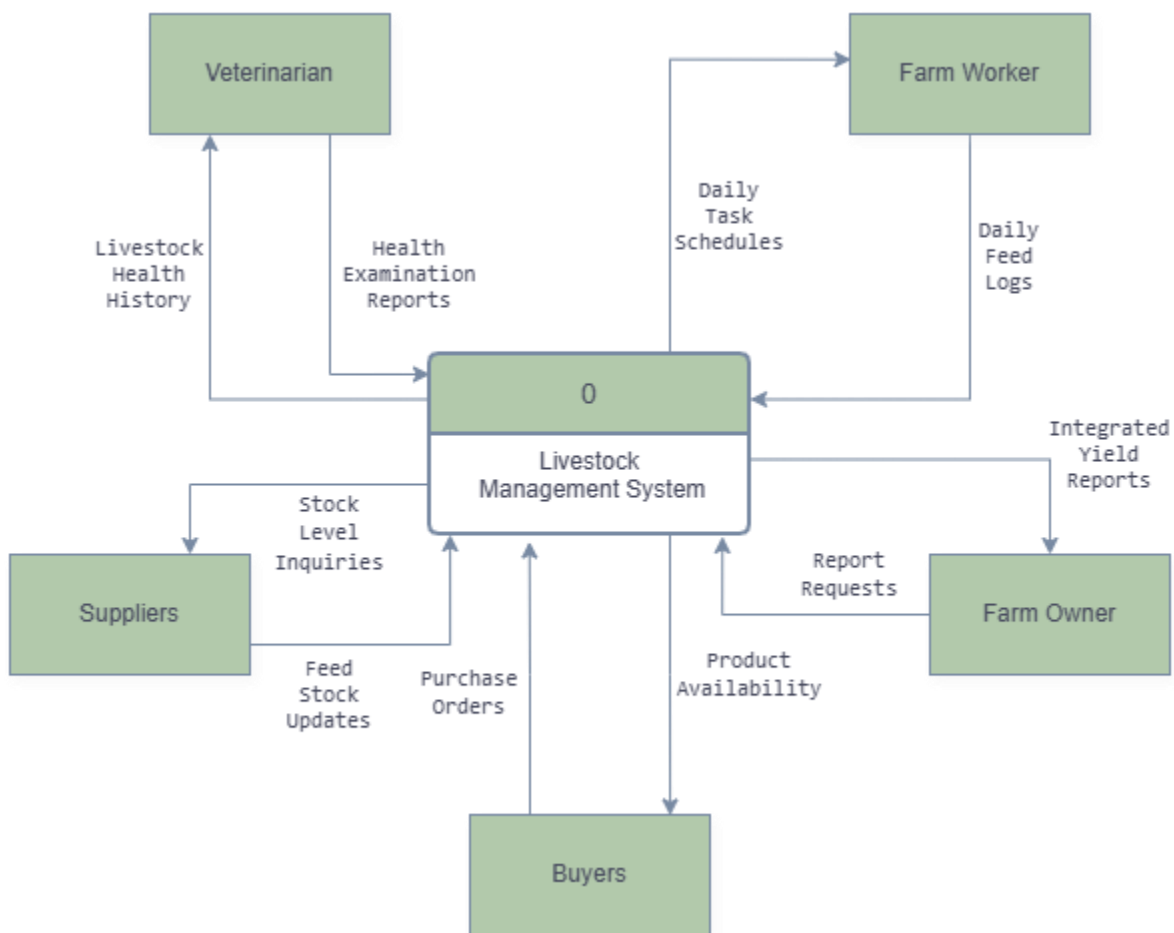


Figure 1: Context-Level Data Flow Diagram — Livestock Management System

External Entity Narrative

The following table describes each external entity and the key data flows between that entity and the Livestock Management System.

External Entity	Role	Inputs to System	Outputs from System
Veterinarian	Licensed animal health professional	Health Examination Reports detailing diagnoses, treatments, and medications administered	Livestock Health History; batch-specific health alerts and treatment schedules
Farm Worker	Operational staff handling day-to-day tasks	Daily Feed Logs (quantities distributed per batch); task completion confirmations	Daily Task Schedules with assigned duties and timing instructions
Farm Owner	Strategic decision-maker and system administrator	Report Requests specifying desired analytics or summaries	Integrated Yield Reports across all three subsystems for performance review
Suppliers	External providers of feed, medicine, and farm supplies	Feed Stock Updates confirming deliveries and stock levels	Stock Level Inquiries requesting replenishment or pricing information

External Entity	Role	Inputs to System	Outputs from System
Buyers / Merchants	Purchasers of livestock products (meat, fish, dairy, eggs)	Purchase Orders from merchants or wholesale buyers	Product Availability notifications indicating current sellable stock

Table 2: External Entity Descriptions and Data Flows

Key Data Flow Summary

The system acts as a central hub connecting five distinct external entities. Inbound data flows primarily consist of operational records (health reports, feed logs, purchase orders, stock updates) while outbound flows deliver processed outputs (health histories, task schedules, yield reports, availability notifications). The Farm Owner interacts bidirectionally, both requesting and consuming summarized reports, making this the most information-intensive relationship at the strategic level.

Preliminary Entity-Relationship (E-R) Diagram

Entity Definitions

The following entities have been identified as the primary data structures required to support the Livestock Management System's core functionalities:

Entity	Definition	Key Attributes
FARM_WORKER	An employee assigned to one or more subsystems responsible for day-to-day operations	WorkerID (PK), Name, Contact Info, Assigned Subsystem
LIVESTOCK_BATCH	A group of animals of the same species managed together through their lifecycle	BatchID (PK), Type, Species, Date of Arrival, Status
HEALTH_RECORD	A documented record of a veterinary examination or treatment event for a batch	RecordID (PK), BatchID (FK), Date, Diagnosis, Treatment Administered
INVENTORY_ITEM	A stockable resource consumed by livestock batches (feed, medicine, supplements)	ItemID (PK), ItemName, Category, Current Stock Level, Unit of Measure
WASTE_LOG	A record of waste materials transferred between subsystems for resource sharing	LogID (PK), TargetSubsystemID, Quantity, Date of Transfer, WorkerID (FK)

Entity	Definition	Key Attributes
SUPPLIER	An external entity providing feed, medicine, or other inventory items	SupplierID (PK), Name, Contact Info, Item Categories Supplied
PURCHASE_ORDER	A transaction record of an order placed by a buyer for farm products	OrderID (PK), BuyerName, ProductType, Quantity, Order Date, Status
TASK_ASSIGNMENT	Specific tasks assigned to workers regarding livestock batches.	TaskID (PK), WorkerID (FK), BatchID (FK), TaskType, ScheduledDate, Status
VETERINARIAN	Medical professionals responsible for examining livestock and submitting records.	VetID (PK), Name, ContactInfo, Specialization
BATCH_INVENTORY	An associative entity tracking the specific consumption of items by batches.	LogID (PK), BatchID (FK), ItemID (FK), QuantityUsed, UsageDate, UsageType

Table 3: Entity Definitions and Key Attributes

E-R Diagram

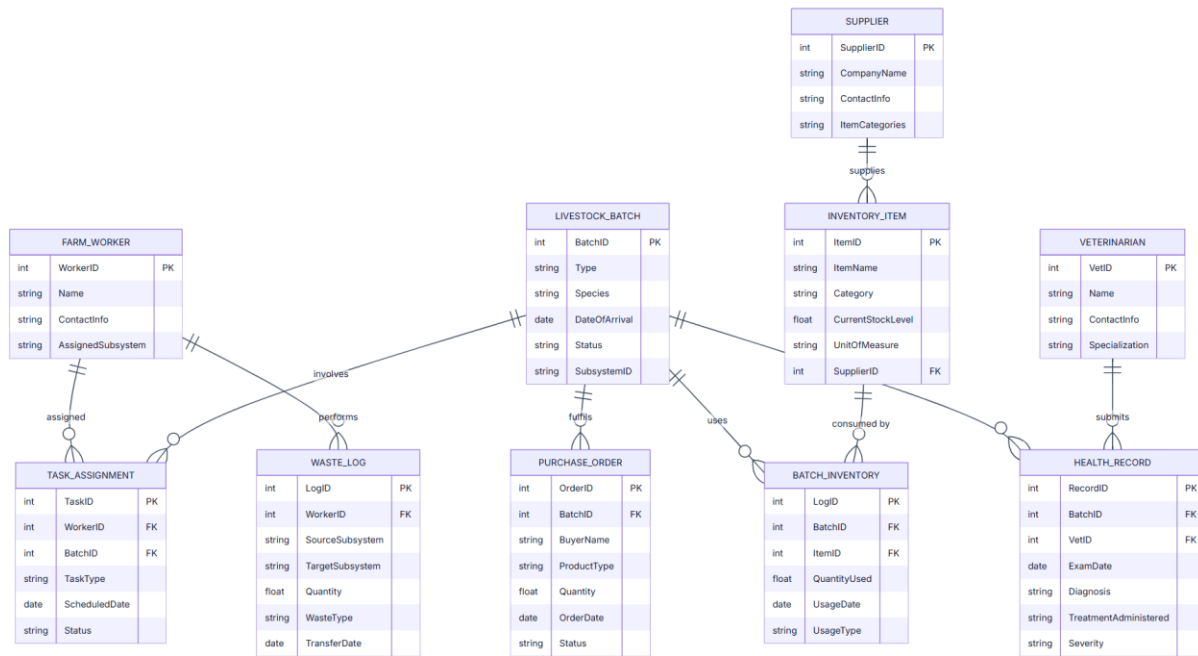


Figure 2: Preliminary Entity-Relationship Diagram — Livestock Management System

Relationship Narrative

The E-R model illustrated establishes a comprehensive operational framework where the **LIVESTOCK_BATCH** serves as the central entity, connecting to **HEALTH_RECORD** in a one-to-many (1:M) relationship to track medical history and to **PURCHASE_ORDER** to manage sales fulfillment. Accountability is maintained through the **FARM_WORKER**, who is linked via 1:M relationships to both **WASTE_LOG** for resource transfers and **TASK_ASSIGNMENT** for daily operations. Resource management is handled through a many-to-many association between **LIVESTOCK_BATCH** and **INVENTORY_ITEM**, which is resolved by the **BATCH_INVENTORY** entity to track specific consumption. Furthermore, **INVENTORY_ITEM** is supported by a 1:M relationship with **SUPPLIER** to ensure stock traceability, while **VETERINARIAN** oversight is integrated through a 1:M relationship with **HEALTH_RECORD**, ensuring professional validation of all clinical treatments administered to the livestock.

High-Level Use Case Diagram

Primary Actors and Their Roles

Actor	Role Description
Farm Owner	Strategic stakeholder with full system access. Manages farm structure, views consolidated financial and yield reports, and initiates long-term planning decisions. Primary consumer of analytics and forecasting features.
Manager	Mid-level operational coordinator responsible for overseeing daily worker activity, allocating resources across subsystems, reviewing performance metrics, and coordinating supplier and merchant relationships.
Farm Worker	Frontline operational user responsible for performing and logging daily tasks including feeding, batch monitoring, and waste handling. Interacts with the system primarily through task schedules and log submission.
Veterinarian	Specialized health professional who accesses batch health records, submits examination reports, records treatment plans, and tracks historical health patterns to support proactive animal care.
Supplier	External entity that receives stock level inquiries from the system and submits feed and supply stock updates to confirm deliveries or adjust inventory records.

Actor	Role Description
Merchant / Buyer	External purchasing entity that submits purchase orders for available farm products and receives product availability notifications from the system.

Table 4: Primary Actors and Their Roles

Use Case List

#	Use Case Name	Description
UC-01	Manage Farm Structure	Farm Owner configures and maintains the organizational structure of the farm including sub-farms, subsystem assignments, and ownership details.
UC-02	Track Livestock Inventory	Owner and Manager monitor current livestock batch counts, species composition, arrival dates, and lifecycle statuses across all subsystems.
UC-03	Process Supplier Orders	Manager or Owner places and tracks feed/supply orders with registered suppliers; system updates inventory upon confirmed delivery.
UC-04	Process Merchant Sales	Manager handles product availability inquiries and processes purchase orders from merchants and buyers.

#	Use Case Name	Description
UC-05	Log Waste Transfers	Farm Worker records inter-subsystem waste transfers (e.g., poultry waste as fish feed), logging quantity, date, and target subsystem.
UC-06	View Financial Overview	Farm Owner reviews profit/loss summaries, cost breakdowns, and revenue trends across all production subsystems.
UC-07	Track Animal Health	Veterinarian records examination results, diagnoses, and treatments for each livestock batch; reviews historical health timelines.
UC-08	Submit Health Reports	Veterinarian submits structured health examination reports linked to specific batch IDs for manager and owner review.
UC-09	Manage Farm Operations	Farm Worker and Manager perform and record day-to-day operational tasks including feeding, cleaning, and batch monitoring.
UC-10	Manage Workers	Manager assigns tasks to workers, monitors activity completion rates, and maintains worker-to-subsystem assignments.
UC-11	Generate Reports	Farm Owner and Manager generate customizable operational, financial, or health reports and export them for external review.

#	Use Case Name	Description
UC-12	Log Daily Feed Activities	Farm Worker records daily feed distribution quantities and schedules per batch, updating inventory depletion records automatically.

Table 5: Use Case List

Use Case Diagram

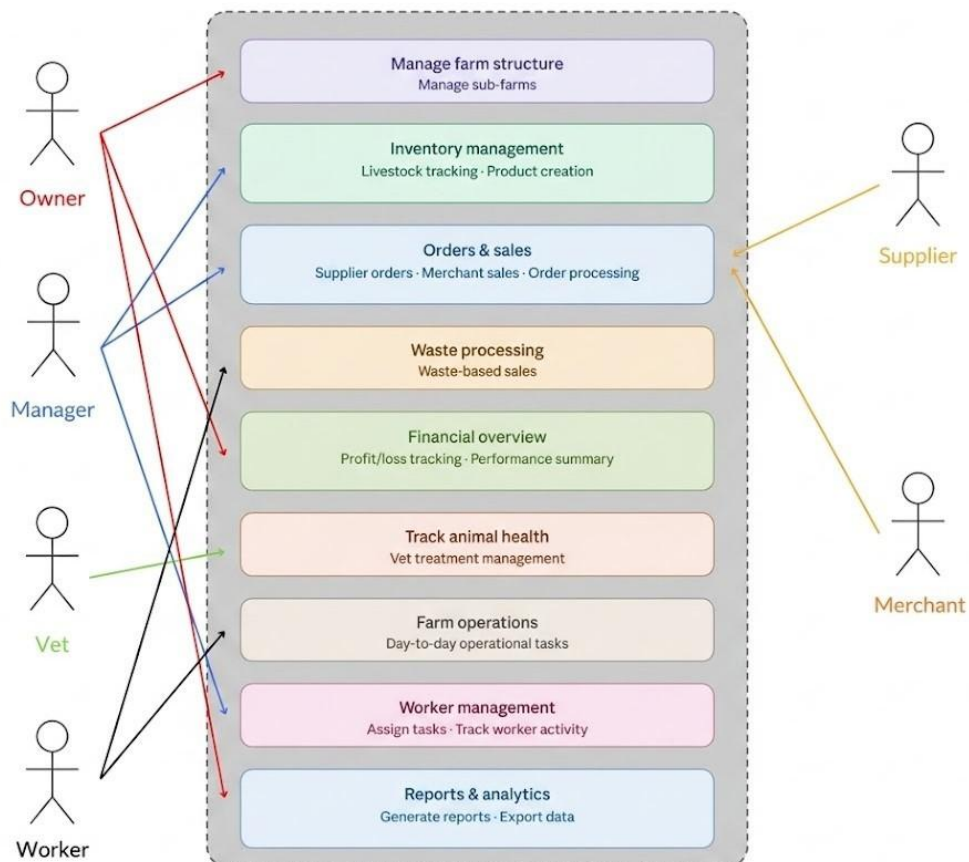


Figure 3: High-Level Use Case Diagram — Livestock Management System

Actor-Use Case Alignment Discussion

The actors and use cases have been carefully selected to reflect the real operational needs and authority boundaries within the integrated farm organization. The Farm Owner is the sole actor interacting with UC-01 (Manage Farm Structure) and UC-06 (Financial Overview) because these use cases directly support strategic, long-horizon decision-making—matching the Farm Owner's role as the top management level. The Veterinarian is restricted to health-related use cases (UC-07, UC-08), ensuring that sensitive clinical data is managed by qualified personnel only, while the Farm Worker is scoped to operational tasks (UC-09, UC-12) that reflect their role as data providers rather than decision-makers.

The Supplier and Merchant actors operate as external entities whose interactions are deliberately limited to transactional use cases (UC-03, UC-04), maintaining system security and data integrity by preventing external parties from accessing internal operational records. This separation supports the security and organizational structure requirements identified during the analysis phase. Together, the twelve use cases comprehensively cover all five core problem areas: UC-12 and UC-05 address manual logging and waste tracking; UC-02 and UC-10 address disconnected subsystems; UC-07 and UC-08 address delayed health decisions; and UC-11 provides the centralized reporting capability currently absent from the organization.

Management Levels and Organizational Culture Analysis

Management Level Identification

The Livestock Management System will be used by or will impact stakeholders across all three classic tiers of organizational management. The table below maps each management level to the relevant organizational roles and their primary information requirements from the system.

Management Level	Organizational Role(s)	Primary Information Requirements	System Features Serving This Level
Operational Control	Farm Workers	Real-time task assignments; daily feed log templates; batch status indicators; shift schedules; simple data entry forms for feed and waste logs	UC-09, UC-12, UC-05; Daily Task Schedules output; Mobile-friendly log entry interface
Managerial Planning & Control	Farm Manager; Veterinarians	Summarized batch health data; short-term inventory forecasts; worker performance summaries; exception alerts (e.g., stock below	UC-07, UC-08, UC-10, UC-03; exception reports; batch-level dashboards

Management Level	Organizational Role(s)	Primary Information Requirements	System Features Serving This Level
		threshold); treatment history per batch	
Strategic Management	Farm Owner	Integrated yield reports across all subsystems; financial profit/loss summaries; what-if forecasting for expansion decisions; trend analysis over time; product availability for sales strategy	UC-01, UC-06, UC-11; consolidated analytics dashboards; long-range forecasting tools

Table 6: Management Levels and Information Requirements

Information Needs by Level

Operational Level — Farm Workers

Farm workers operate at the frontline of daily production and require a system interface that minimizes cognitive overhead. Their primary information needs are highly structured and repetitive: they need to know what tasks to perform each shift, how much feed to distribute to each batch, and how to record waste transfers correctly. The system must provide pre-populated forms, clear batch identifiers, and instant confirmation of log submission. Real-time data is

critical at this level—stale information about batch status or feed schedules could result in animal health risks or resource waste.

Managerial Level — Farm Manager and Veterinarians

The managerial tier requires both real-time visibility and historical trend access. The Farm Manager needs to allocate workers efficiently across the three subsystems, monitor batch progress against targets, and respond quickly to deviations such as low inventory levels or declining yield metrics. Veterinarians, functioning as a specialized managerial subgroup, primarily need structured access to batch health timelines, treatment records, and examination submission workflows. Both roles benefit from exception-reporting functionality that surfaces anomalies without requiring exhaustive manual review of all records.

Strategic Level — Farm Owner

The Farm Owner operates with the broadest information horizon and the lowest tolerance for operational detail. Strategic decisions—such as expanding the fisheries pond, introducing a biogas plant, or entering new distribution markets—require aggregated, cross-subsystem data presented in summary dashboards. The Farm Owner's most valued system feature will be the integrated yield report and what-if scenario modelling capability, as these directly support investment and expansion decisions. External environmental data (market pricing, disease outbreak alerts) could further enrich strategic decision-making in future system iterations.

Organizational Culture Analysis

The organizational culture of the integrated farm is best characterized as a traditional, task-oriented culture with strong operational norms built around manual, paper-based processes. This culture has developed over years of habitual practice and is reinforced by informal hierarchies, where senior workers and long-tenured staff hold significant informal authority over how daily operations are carried out. The farm's cultural identity is closely tied to tangible, hands-on

interaction with the animals and environment, rather than digital data entry or screen-based monitoring.

Current Cultural State

The prevailing subculture among farm workers is one of procedural familiarity: tasks are performed according to established routines, and any deviation—including the introduction of a digital system—may be perceived as a disruption to a known, working rhythm. Workers may view data entry as an additional burden rather than a productivity enabler. Furthermore, veterinary staff often maintain their own informal records alongside any mandated documentation, reflecting a subculture that values professional autonomy and personal judgment over centralized data governance.

Expected System Impact and Resistance

The primary cultural challenge for the Livestock Management System will be resistance to change from farm workers. This resistance is anticipated to manifest in several ways: reluctance to replace paper logs with digital submissions, skepticism about the accuracy or usefulness of computer-generated schedules, and distrust of data-driven recommendations that contradict established practices. The managerial subculture may face its own tension between embracing the efficiency gains of real-time dashboards and the perceived loss of discretionary authority when system alerts preempt managerial judgment.

Mitigation Strategy

To address these cultural barriers, the system design must prioritize an extremely intuitive, mobile-friendly interface for farm workers, minimizing the learning curve and reducing the perceived complexity of digital logging. Structured, role-based training programs will be essential to build confidence and demonstrate the personal benefits of the system to each user group. For the Farm Owner and Manager, early demonstrations of the integrated reporting dashboard—showing insights that were previously impossible with manual methods—will help

establish trust in data-driven decision-making. A phased rollout strategy, beginning with the highest-impact and most receptive user group, is recommended to build organizational momentum and reduce the perceived risk of wholesale adoption.

Appendix

Term	Definition
Livestock Batch	A group of animals of the same species managed collectively through a shared lifecycle within one subsystem
Subsystem	One of the three primary farm units: Poultry, Fisheries, or Cattle
Waste Transfer	The process of moving organic by-products (e.g., poultry droppings) from one subsystem to another for use as resource input
Context-Level DFD	A data flow diagram showing the system as a single process with all external entities and major data flows
E-R Diagram	Entity-Relationship Diagram; a graphical model representing the data entities, their attributes, and the relationships between them
Cardinality	In an E-R diagram, the numerical constraint expressing how many instances of one entity relate to instances of another (1:1, 1:M, M:N)
Use Case	A description of how an actor achieves a specific goal through interaction with the system
Actor	A person, system, or external entity that interacts with the system from outside its boundary
Operational Control	The lowest tier of management focused on executing structured, day-to-day operational tasks

Term	Definition
Strategic Management	The highest tier of management focused on long-term organizational direction and goal-setting

Table A1: Glossary of Key Terms